

Portfolio Resilience: Backtesting Asset Allocation Strategies During Financial Crises

DATA 510 M1 Practice Proposal

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1. Title and Team Metadata

Project title: Portfolio Resilience: Backtesting Asset Allocation Strategies During Financial Crises

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Team size: 2

Peer Stakeholder POs: Ben Dyer, Luca Casano, Manish Reddy Kallu

Studio Session: 4

2. Problem Framing and Research Questions

2.1 Background: Portfolio construction under market stress

Many stakeholders evaluate portfolios based on expected returns, recent performance, or exposure to popular sectors. However, financial crises often reveal risks that are not obvious during normal market conditions. Portfolios that appear strong during growth years may become fragile when volatility rises, correlations shift, and drawdowns accelerate.

This project focuses on backtesting different portfolios across periods of financial stress. The goal is to evaluate how different sectors and asset classes behave during crisis periods compared to normal market conditions. The project will compare return potential, downside protection, volatility, recovery behavior, and risk-adjusted performance across multiple portfolio strategies.

2.2 Why this project matters to our team

Our team is interested in finance, investment decision-making, and quantitative analysis. We want to move beyond simple return comparisons and create a more risk-aware way to evaluate returns on appreciating assets.

This project supports our team vision: **to create an innovative lens through which stakeholders can view risk and portfolio construction.** By comparing portfolios across crisis and growth periods, we can better understand which allocations are more fragile, which are more resilient, and whether downside protection comes at the cost of lower gains during strong market years.

2.3 Problem statement

Problem: Stakeholders often construct or evaluate portfolios based on expected gains without fully understanding how those portfolios behave during financial crises and periods of market uncertainty.

Consequence if unsolved: Stakeholders may overestimate portfolio strength during normal markets and underestimate exposure to drawdowns, volatility, sector concentration, and delayed recovery during market stress.

What success would change this term: This project will produce a reproducible backtesting workflow, a set of portfolio performance comparisons, and clear evidence on how different portfolio constructions perform across crisis and non-crisis periods.

2.4 Research questions

Primary research question:

How can portfolio optimization help stakeholders construct portfolios that preserve return potential while reducing drawdowns during periods of financial crisis and market uncertainty?

Sub questions:

1. Do drawdown-resistant portfolios experience significantly lower gains during growth years compared to regular market indexes?
2. How does portfolio performance during crisis periods compare to performance during normal market conditions?
3. How are Sharpe and Sortino ratios affected during periods of uncertainty compared to periods of growth?
4. How do different asset classes and sectors, such as technology, energy, real estate, consumer staples, and defensive treasury-based assets, behave during periods of market stress?

3. Impact and Stakeholders

3.1 Who benefits

The main stakeholders include:

- Investors
- Portfolio managers
- Financial analysts
- Finance educators
- Investment clubs
- Risk managers
- Students studying finance or portfolio analytics

These stakeholders may benefit from a clearer framework for evaluating portfolio construction under market stress. Instead of only asking which portfolio produces the highest return, this project asks which portfolios preserve value, limit drawdowns, and recover more effectively during uncertainty.

3.2 Practical impact

If successful, this project will help stakeholders understand how portfolio construction choices affect performance during both growth periods and crisis periods. The project will show how drawdowns, volatility, recovery time, Sharpe ratio, and Sortino ratio change across different market environments.

The project is not meant to provide investment advice. Its value is educational and analytical. It will show how historical backtesting can help stakeholders think more critically about risk, diversification, asset allocation, and portfolio resilience.

4. Data Sources and Access Plan

Source	Description	Access	Expected Volume	Refresh	Approval Status
Yahoo Finance through yfinance	Historical adjusted price data for stocks, ETFs, and benchmarks	Public API wrapper	Daily prices from 2007 to 2026 for selected assets	Daily market updates	<i>Pending instructor approval</i>
FRED	Risk-free rate, interest rates, inflation, and macroeconomic context	Public API/download	Daily, monthly, or quarterly macro series	Varies by series	<i>Pending instructor approval</i>
Portfolio construction table	Team-defined asset allocations and weights	Created by team	Small structured table	Static unless revised	Team-created
Crisis period definitions	Dates for financial stress periods	Team-defined using historical market periods	Small reference table	Static	Team-created
Fama-French composite market index (Rm-Rf)	Rm-Rf includes all NYSE, AMEX, and NASDAQ firms.	Fama-French database	Monthly Prices going back to 1929	Monthly market updates	<i>Pending instructor approval</i>

4.1 Likely assets and proxies

The project will use a combination of individual equities, sector ETFs, and benchmark ETFs. Potential asset groups include:

- Technology: Apple, Microsoft, Nvidia, Amazon, Google, Meta, Broadcom, Tesla
- Energy: ExxonMobil, Chevron, ConocoPhillips, EOG Resources, Schlumberger, utilities, and energy-related holdings
- Real estate: VNQ or similar REIT ETF
- Consumer staples: Walmart, Costco, Target, Procter & Gamble, Coca-Cola, PepsiCo, General Mills, and similar defensive consumer names

- Defensive assets: Treasury ETF or bond ETF
- Benchmarks: SPY, QQQ, BND, Fama-French Rm-Rf market composite, and a 60/40 portfolio

4.2 Data limitations

Some assets may not have complete price history back to 2007. If an asset lacks enough history, we will either shorten the comparison period, use an ETF proxy, or clearly document the limitation. Volatility indexes such as VIX are not directly investable, so if used, they will be treated as stress indicators or replaced with investable proxies where appropriate.

5. Data Engineering Plan

The project will use a reproducible data pipeline organized around the DS3-style structure.

Planned folder structure:

```
data/
  raw/
    market_prices/
    fred_macro/
  interim/
    cleaned_prices/
    tidy_returns/
  processed/
    portfolio_returns/
    crisis_period_metrics/

notebooks/
  01_data_ingestion.ipynb
  02_data_validation.ipynb
  03_eda_backtesting.ipynb
  04_crisis_analysis.ipynb
  05_portfolio_optimization.ipynb

src/
  data/
    pull_market_data.py
    clean_market_data.py
  analysis/
    performance_metrics.py
    drawdown_analysis.py
    optimization.py
```

5.1 Data pipeline

The pipeline will:

1. Pull adjusted historical price data.
2. Convert prices into daily returns.
3. Join returns to portfolio weights.
4. Calculate portfolio-level returns.
5. Label observations as crisis or normal market periods.
6. Calculate performance metrics by portfolio and period.
7. Export processed tables for analysis and visualization.

5.2 Data quality checks

Quality checks will include:

- Missing price data by ticker
- First and last available trading date
- Duplicate dates
- Assets with incomplete history
- Portfolio weights summing to 1
- Crisis periods with enough observations for comparison

6. Planned Analysis

6.1 Descriptive EDA

The first phase will explore the data and summarize portfolio behavior.

Planned EDA outputs:

- Cumulative return charts by portfolio
- Drawdown charts by portfolio
- Summary table of total return, volatility, Sharpe ratio, Sortino ratio, and maximum drawdown
- Correlation matrix across portfolios and benchmarks
- Crisis vs normal market comparison

6.2 Crisis period analysis

The project will compare portfolios across selected financial stress periods.

Crisis Period	Approximate Window
Global Financial Crisis	2007 to 2009
COVID crash	February 2020 to April 2020
2022 inflation and rate-hike shock	2022
Regional banking crisis	March 2023 to May 2023

For each crisis period, the project will calculate:

- Total return
- Maximum drawdown
- Annualized volatility
- Sharpe ratio
- Sortino ratio
- Worst daily return
- Recovery time where feasible

6.3 Portfolio optimization

After evaluating the initial portfolio constructions, the project will test whether optimized portfolios can improve risk-adjusted performance. Optimization may include:

- Maximum Sharpe portfolio
- Minimum volatility portfolio
- Minimum drawdown portfolio
- Balanced return and drawdown-resistant portfolio

These optimized portfolios will be compared against the original constructed portfolios and benchmarks.

6.4 Evaluation plan

The project will answer the research question by comparing portfolios on:

- Return potential: total return and annualized return
- Downside protection: maximum drawdown and worst daily return
- Risk-adjusted performance: Sharpe ratio and Sortino ratio
- Stability: volatility and crisis-period performance
- Opportunity cost: performance during growth years compared to SPY, QQQ, and a 60/40 benchmark

A portfolio will be considered more resilient if it has lower drawdowns, lower volatility, stronger Sortino ratio, and reasonable return preservation compared to benchmarks.

7. Program Integration: Five Pillars

Pillar	Plan
Data engineering	Build a reproducible pipeline for downloading, cleaning, and transforming market data into portfolio return tables.
Analytics and machine learning	Use quantitative backtesting and portfolio optimization to compare risk and return across strategies.
Visualization and communication	Create cumulative return charts, drawdown charts, crisis comparison visuals, and risk-return plots.
Ethics and responsible use	Clearly state that the analysis is educational and not investment advice. Avoid overstating what historical backtests can prove.
Statistics and research design	Use defined crisis windows, benchmark comparisons, risk-adjusted metrics, and clear evaluation criteria.

8. Ethics and Risk Notes

8.1 Investment advice risk

This project could be misread as recommending specific investments. To avoid that, all results will be framed as historical analysis, not financial advice. The project will clearly state that past performance does not guarantee future results.

8.2 Backtesting bias

Backtesting can create false confidence if portfolios are built using knowledge of what performed well historically. We will address this by documenting portfolio construction rules before evaluation and avoiding claims that the results will hold in future crises.

8.3 Data quality and survivorship bias

Some assets may have shorter histories or may not represent companies that failed or disappeared. This creates survivorship bias. Where possible, the project will use broad ETFs and benchmarks to reduce this issue.

8.4 Fair comparison

Different assets have different start dates. The project will either use a common comparison period or clearly state when portfolios are compared over different available windows.

9. Timeline and Milestones

Week	Plan
Week 4	Submit M1 proposal, finalize research questions, define portfolios and benchmarks
Week 5	Pull historical market data, build raw and interim datasets, validate missingness and start dates
Week 6	Complete EDA, cumulative returns, drawdowns, and initial performance metrics
Week 7	Submit M2 data summary, finalize crisis windows and benchmark comparison
Week 8	Complete crisis-period analysis and normal vs crisis market comparison
Week 9	Build first optimization models and compare against constructed portfolios
Week 10	Submit M3 poster draft with early findings and visualizations
Week 11	Refine analysis based on peer and instructor feedback
Week 12	Submit M4 write-up draft, finalize tables and charts
Week 13	Rehearse presentation and polish dashboard or notebook outputs
Week 14	Submit M5 final poster, write-up, and project repository

Risks and mitigations

Risk	Mitigation
Some assets lack enough history	Use ETF proxies or restrict analysis to common date ranges
VIX is not directly investable	Treat it as a stress proxy or replace with tradable volatility ETF
Optimization becomes too broad	Limit optimization to a small set of broad asset ETFs
Backtest results are overinterpreted	Frame findings as historical and educational only
Too many portfolios create scope creep	Focus on five core portfolio constructions and three to four benchmarks

10. DS3 Process and Tooling

- **GitHub repo:** <https://github.com/awjackson00/data510--Quant-Back-Testing-Projects>
- **Projects board:** <https://github.com/users/awjackson00/projects/2>
Discord: Project 4: Alex & Dane
- **Vision:** To create an innovative lens through which stakeholders can view risk and portfolio construction.
- **Mission:** To build a reproducible portfolio backtesting and optimization workflow that evaluates how different asset allocations perform during financial crises and normal market conditions, using risk-adjusted metrics such as drawdown, volatility, Sharpe ratio, Sortino ratio, and recovery behavior.
- **CHARTER.md alignment:** The proposal aligns with the team vision of creating an innovative lens for viewing risk and portfolio construction. The project focuses on portfolio optimization, drawdown resistance, and crisis-period performance.

10.1 Initial backlog items

PBI

Create

Observe

Analyze

Define portfolio reference table	Create table of portfolios, assets, tickers, and weights	Confirm weights sum to 1 for each portfolio	Evaluate whether portfolios represent distinct risk strategies
Pull historical prices	Download adjusted prices for all assets and benchmarks	Check missing values and asset start dates	Decide whether to use common date window or asset-specific windows
Build portfolio returns	Convert prices to returns and apply weights	Confirm portfolio return series are created	Compare early performance across portfolios
Analyze crisis periods	Label crisis and normal market periods	Calculate crisis-period metrics	Identify which portfolios are most fragile or resilient
Optimize portfolio weights	Simulate or optimize portfolio allocations	Compare optimized portfolio to benchmarks	Assess whether optimization improves drawdown resistance and return preservation

11. Appendix: References and Data Links

1. Yahoo Finance / [yfinance](https://pypi.org/project/yfinance/) documentation:
<https://pypi.org/project/yfinance/>
2. Federal Reserve Economic Data, FRED: <https://fred.stlouisfed.org/>
3. SPDR S&P 500 ETF Trust, SPY:
<https://www.ssga.com/us/en/intermediary/etfs/funds/spdr-sp-500-etf-trust-spy>
4. Invesco QQQ ETF: <https://www.invesco.com/qqq-etf/en/home.html>
5. Vanguard Real Estate ETF, VNQ:
<https://investor.vanguard.com/investment-products/etfs/profile/vnq>
6. Vanguard Total Bond Market ETF, BND:
<https://investor.vanguard.com/investment-products/etfs/profile/bnd>